

University of Sri Jayewardenepura

Department of Mathematics

MAT 2011

Linear Algebra II

Semester I 2026

Exercise Sheet 1

Due: June 11th, 2026 by 12noon

In problems 2-8 assume that the matrix A has the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$.

1. Prove that “A square matrix is invertible if and only if its determinant is non-zero”.
2. By using the theorem in problem 1, prove that “A square matrix A is invertible if and only if $\lambda=0$ is not an eigenvalue of A ”.
3. Calculate eigenvalues and eigenspaces of the given matrix. If the algebraic multiplicity of an eigenvalue is greater than 1, calculate the geometric multiplicity.

I. $\begin{pmatrix} -12 & 7 \\ -7 & 2 \end{pmatrix}$

II. $\begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix}$

III. $\begin{pmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{pmatrix}$

IV. $\begin{pmatrix} a & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & a \end{pmatrix}$

V. $\begin{pmatrix} a & b & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & a \end{pmatrix}, b \neq 0$

VI. $\begin{pmatrix} a & b & 0 & 0 \\ 0 & a & c & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & a \end{pmatrix}, bc \neq 0$

VII. $\begin{pmatrix} a & b & 0 & 0 \\ 0 & a & c & 0 \\ 0 & 0 & a & d \\ 0 & 0 & 0 & a \end{pmatrix}, bcd \neq 0$

4. Show that the eigenvalues of A^T are $\lambda_1, \lambda_2, \dots, \lambda_k$.
5. Show that the eigenvalues of αA are $\alpha\lambda_1, \alpha\lambda_2, \dots, \alpha\lambda_k$.
6. Show that A^{-1} exists if and only if $\lambda_1\lambda_2 \dots \lambda_k \neq 0$.
7. If A^{-1} exists, show that the eigenvalues of A^{-1} are $1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_k$.

8. Show that the matrix $A - \alpha I$ has the eigenvalues $\lambda_1 - \alpha, \lambda_2 - \alpha, \dots, \lambda_k - \alpha$.
9. Show that the eigenvalues of A^2 are $\lambda_1^2, \lambda_2^2, \dots, \lambda_k^2$.
10. Show that the eigenvalues of A^m are $\lambda_1^m, \lambda_2^m, \dots, \lambda_k^m$ for $m=1,2,3,\dots$
11. Let λ be an eigenvalue of A with corresponding eigenvector $\mathbf{v}$. Let $p(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \dots + a_n\lambda^n$. Define the matrix $p(A) = a_0I + a_1A + a_2A^2 + \dots + a_nA^n$. Show that $p(A)\mathbf{v} = p(\lambda)\mathbf{v}$.
12. Using the result of problem 9, show that if $\lambda_1, \lambda_2, \dots, \lambda_k$ are eigenvalues of A , then $p(\lambda_1), p(\lambda_2), \dots, p(\lambda_k)$ are eigenvalues of $p(A)$.

13. Let $A_1 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$, $A_2 = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$, $A_3 = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$ and

$A_4 = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$ Show that, for each matrix, $\lambda=2$ is an eigenvalue of algebraic

multiplicity 4. In each case compute the geometric multiplicity of $\lambda=2$.



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Reference: Elementary Linear Algebra by Grossman.